The physical meaning of transverse / longitudinal fields

$$\begin{cases} \vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \\ \vec{\nabla}\cdot\vec{B} = 0 \\ \vec{\nabla}\times\vec{E} + \frac{\partial\vec{B}}{\partial t} = 0 \\ \vec{\nabla}\times\vec{B} - \frac{1}{c^2}\frac{\partial\vec{E}}{\partial t} = \mu_0\vec{j} \end{cases}$$

Act with $\vec{\nabla}\cdot$ on last eq $\Rightarrow \vec{\nabla}\cdot(\vec{\nabla}\times\vec{B}) = 0$

$$\Rightarrow \quad -\varepsilon_0\mu_0\frac{\partial}{\partial t}\underbrace{\vec{\nabla}\cdot\vec{E}}_{\frac{\rho}{\varepsilon_0}} = \mu_0\vec{\nabla}\cdot\vec{j} \Rightarrow \left[\frac{\partial\rho}{\partial t} + \vec{\nabla}\cdot\vec{j} = 0\right] \text{ continuity eq}$$

$$\begin{cases} \vec{E} = -\vec{\nabla}\varphi - \frac{\partial\vec{A}}{\partial t} \\ \vec{B} = \vec{\nabla}\times\vec{A} \end{cases}$$

Gauge freedom $\begin{cases} \varphi \to \varphi' = \varphi - \frac{\partial\chi}{\partial t} \\ \vec{A} \to \vec{A}' = \vec{A} + \vec{\nabla}\chi \end{cases} \Rightarrow \begin{cases} \vec{E} \to \vec{E}' = -\vec{\nabla}\varphi + \overbrace{\frac{\partial}{\partial t}\vec{\nabla}\chi - \frac{\partial}{\partial t}\vec{\nabla}\chi}^{0} \\ \vec{B} \to \vec{B}' = \vec{\nabla}\times\vec{A} + \underbrace{\vec{\nabla}\times\vec{\nabla}\chi}_{0} \end{cases}$

the physical fields do not change

Every vector field can be separated in longitudinal (zero curl) and transverse (zero divergence) components: $\vec{E} = \vec{E}^L + \vec{E}^T$

such that $\vec{\nabla}\times\vec{E}^L = 0$ and $\vec{\nabla}\cdot\vec{E}^T = 0$

Provided that $\vec{E}$ vanishes at infinity (boundary condition), the separation is <u>exact</u> and <u>unique</u>.

Maxwell's eqns:

$\vec{\nabla}\cdot\vec{B}^L = 0 \Rightarrow \vec{B}^L = 0$ everywhere $\Rightarrow \vec{B} = \vec{B}^T$ is purely transverse.

From $\vec{\nabla}\cdot\vec{E}^L = \frac{\rho}{\varepsilon_0}$ we know that longitudinal electric

field is determined by charge density ρ.

Moreover, $\vec{\nabla}\cdot\frac{\partial \vec{E}^L}{\partial t} = \frac{1}{\varepsilon_0}\frac{\partial \rho}{\partial t} = -\frac{1}{\varepsilon_0}\vec{\nabla}\cdot\vec{j} = -\frac{1}{\varepsilon_0}\vec{\nabla}\cdot\vec{j}^L$

continuity eqn

therefore $\frac{\partial \vec{E}^L}{\partial t} = -\frac{1}{\varepsilon_0}\vec{j}^L$ (since curl of any longitudinal field is zero!)

$$\Rightarrow \vec{\nabla}\times\vec{B}^T - \varepsilon_0\mu_0\frac{\partial}{\partial t}\left(\vec{E}^L+\vec{E}^T\right) = \mu_0\left(\vec{j}^L+\vec{j}^T\right)$$

$$\left.\begin{aligned} &\Rightarrow \vec{\nabla}\times\vec{B}^T - \varepsilon_0\mu_0\frac{\partial \vec{E}^T}{\partial t} = \mu_0\vec{j}^T \\ &\vec{\nabla}\times\vec{E}^T + \frac{\partial \vec{B}}{\partial t} = 0 \Rightarrow \vec{\nabla}\times\vec{E}^T + \frac{\partial \vec{B}^T}{\partial t} = 0 \end{aligned}\right\} \text{transverse fields only}$$

$$\left.\vec{\nabla}\cdot\vec{B}^L = 0, \quad \vec{\nabla}\cdot\vec{E}^L = \frac{\rho}{\varepsilon_0}\right\} \text{longitudinal only}$$

Finally, gauge transformation affects only longitudinal part of $\vec{A}$.

$$\varphi \to \varphi' = \varphi - \frac{\partial}{\partial t}\chi$$
$$\vec{A}^L \to \vec{A}^{L'} + \vec{\nabla}\chi$$
$$\vec{A}^T \to \vec{A}^{T'} = \vec{A}^T$$

Now we Fourier transform: $i\vec{q}\cdot\vec{E}^L(q,\omega) = \frac{\rho}{\varepsilon_0}$

we also know $\vec{\nabla}\times\vec{E}^L = 0 \Rightarrow \vec{q}\times\vec{E}^L(q,\omega) = 0 \Rightarrow \vec{E}^L = \vec{q}\,f$

$$\Rightarrow i\vec{q}\cdot\vec{q}\,f(q,\omega) = \frac{\rho(q,\omega)}{\varepsilon_0} = i\vec{q}\cdot\vec{E}^L = i\vec{q}\cdot\vec{E}$$

since $\vec{q}\cdot\vec{E}^T = 0$

$$\Rightarrow f(q,\omega) = -i\frac{\rho(q,\omega)}{q^2} \Rightarrow \vec{E}^L = -i\frac{\vec{q}\,\rho(q,\omega)}{q^2}$$

OR $f = \frac{\vec{q}\cdot\vec{E}}{q^2} \Rightarrow \vec{E}^L = \frac{\vec{q}\left(\vec{q}\cdot\vec{E}\right)}{q^2}$

in components: $E^L_a = \frac{q_a q_b}{q^2}E_b = P^L_{ab}E_b$

longitudinal projector

now $\vec{E}^T = \vec{E} - \vec{E}^L = \vec{E} - \frac{(\vec{q}\cdot\vec{E})\vec{q}}{q^2}$

$$\Rightarrow E^T_a = \left(\delta_{ab} - \frac{q_a q_b}{q^2}\right) E_b = P^T_{ab} E_b$$

transverse projector

We observe that $P^T_{ab} + P^L_{ab} = \delta_{ab}$

in matrix notations: $\hat{P}^T + \hat{P}^L = \hat{1}$

expresses completeness of longitudinal – transverse separation.

Observe that $P^L_{ab} P^L_{bc} = \frac{q_a q_b}{q^2}\frac{q_b q_c}{q^2} = q^2 \frac{q_a q_c}{q^4} = \frac{q_a q_c}{q^2} = P^L_{ac}$

$$P^T_{ab} P^L_{bc} = \left(\delta_{ab} - \frac{q_a q_b}{q^2}\right)\frac{q_b q_c}{q^2} = \delta_{ab}\frac{q_b q_c}{q^2} - \frac{q_a q_c}{q^2} = 0$$

$$P^T_{ab} P^T_{bc} = \left(\delta_{ab} - \frac{q_a q_b}{q^2}\right)\left(\delta_{bc} - \frac{q_b q_c}{q^2}\right) = \delta_{ab}\delta_{bc} - \frac{q_a q_b}{q^2}\delta_{bc}$$

$$= \delta_{ac} - \frac{q_a q_c}{q^2} = P^T_{ac}$$

Thus projectors $P^{T,L}$ are orthogonal to each other and are idempotent: $(P^{T,L})^2 = P^{T,L}$.

Therefore Ohm's law in the isotropic medium:

$$\vec{j}(r,t) = \int_{-\infty}^{t} dt' \int d\vec{r}'\, \sigma(\vec{r}-\vec{r}', t-t')\, \vec{E}(\vec{r}', t')$$

implies $\vec{j}_a(q,\omega) = \sigma_{ab}(q,\omega) E_b(q,\omega)$

where $\sigma_{ab} = \frac{q_a q_b}{q^2}\sigma^L(q,\omega) + \left(\delta_{ab} - \frac{q_a q_b}{q^2}\right)\sigma^T(q,\omega)$

$$\Rightarrow \vec{j}^{L,T}(q,\omega) = \sigma^{L,T}(q,\omega)\, \vec{E}^{L,T}(q,\omega)$$

Note that continuity eq involves only $\vec{j}^L$: $\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot\vec{j}^L = 0$

$$\Rightarrow -i\omega\rho(q,\omega) + i\vec{q}\cdot\vec{j}^L = 0 \Rightarrow q j^L(q,\omega) = \omega\rho(q,\omega).$$

$\Rightarrow$ Transverse current $\vec{j}^T$ does not lead to accumulation of charge.

This separation of conductivity is fully consistent with result for dielectric function we derived previously:

$$\tilde{\varepsilon}_{\alpha\beta}(k,\omega) = \left(\delta_{\alpha\beta} - \frac{k_\alpha k_\beta}{k^2}\right)\tilde{\varepsilon}_{tr} + \frac{k_\alpha k_\beta}{k^2}\tilde{\varepsilon}_{\ell}$$

Since $\varepsilon = \varepsilon_0 + \dfrac{i\,\sigma(k,\omega)}{\omega}$

Thus $\varepsilon_{tr,\ell}$ are related to $\sigma^{T,L}$.